

DSA BASIC TEST — BATCH B

Duration: 1 Hour 30 Minutes (90 Minutes) | Total Questions: 30

Instructions

1. Test duration: 1 hour 30 minutes (90 minutes).
2. Attempt all 30 questions.
3. This is an individual test. Do not cheat, copy another student's work, or use unauthorized help.
4. Do not use ChatGPT, Gemini, Copilot, search engines, or other AI/online solution sources during the test unless the instructor explicitly permits them.
5. Write clean, readable code and use meaningful variable names.
6. For coding questions, include the complete program or function as requested.
7. After completing the test, create a GitHub repository and push your complete test work to it.
8. Use a clear repository name such as DSA_Test_Batch_A or DSA_Test_Batch_B.
9. Submit the GitHub repository link to the instructor before the deadline.
10. Do not copy another student's repository or code. Your submission must be your own work.

BASIC DSA — THEORY

1. What is an algorithm? List two characteristics of a good algorithm.

Answer: _____

2. Explain the difference between static and dynamic data structures.

Answer: _____

3. What is an abstract data type (ADT)? Give one example.

Answer: _____

4. What is the difference between a stack's push and pop operations?

Answer: _____

5. What is a queue's enqueue operation? What happens during dequeue?

Answer: _____

6. What is the best-case and worst-case time complexity of linear search?

Answer: _____

7. Why must data generally be sorted before applying binary search?

Answer: _____

8. What is a singly linked list? How is it different from an array?

Answer: _____

9. What is a binary tree? What is the maximum number of children of a node?

Answer: _____

10. Differentiate between BFS and DFS at a basic conceptual level.

Answer: _____

PATTERN PRINTING

11. Write a C/C++ program to print a left-aligned inverted triangle of stars for N rows.

Code/Answer: _____

12. Write a C/C++ program to print this pattern for N=5:

1
22
333
4444
55555

Code/Answer: _____

13. Write a C/C++ program to print a rectangular pattern of N rows and M columns using stars.

Code/Answer: _____

14. Write a C/C++ program to print a centered pyramid of stars for N rows.

Code/Answer: _____

15. Write a C/C++ program to print the number pattern: 12345, 1234, 123, 12, 1.

Code/Answer: _____

CODING QUESTIONS

16. Write a program to calculate the sum of the first N natural numbers.

Code/Answer: _____

17. Write a program to check whether a given number is an Armstrong number.

Code/Answer: _____

18. Write a program to find the GCD of two positive integers.

Code/Answer: _____

19. Write a program to find the LCM of two positive integers.

Code/Answer: _____

20. Write a program to count the frequency of a given element in an array.

Code/Answer: _____

21. Write a program to find the smallest element in an array.

Code/Answer: _____

22. Write a program to reverse an array without using another array.

Code/Answer: _____

23. Write a program to remove duplicate elements from a simple integer array.

Code/Answer: _____

24. Write a program to merge two arrays into a third array.

Code/Answer: _____

25. Write a program to check whether an integer is a perfect number.

Code/Answer: _____

26. Write a program to print the multiplication table of a given number.

Code/Answer: _____

27. Write a program to count vowels and consonants in a string.

Code/Answer: _____

28. Write a program to implement a queue using an array with enqueue and dequeue operations.

Code/Answer: _____

29. Write a program to insert a node at the beginning of a singly linked list.

Code/Answer: _____

30. Write a program to search for a value in a singly linked list.

Code/Answer: _____